

KidsLearn — Frontend (UI) qurish rejasi

1–7 yoshdagi bolalar uchun interaktiv ta'lim platformasi. Uchta rol, uchta tajriba: administrator paneli, ota-ona kabineti, bola rejimi.

Backend holati: tayyor va production darajasida — 13 controller, 14 servis, `yarn typecheck` toza o'tadi, 4 ta migratsiya.

Frontend holati: boshqa loyihadan (EduTest) qolgan skelet — noldan quriladi.

Reja hajmi: 13 bosqich (0–12), har biri mustaqil tekshiriladigan.

Mundarija

- Loyiha konteksti va nima saqlanadi
- Texnologiyalar to'plami
- API audit natijasi (spec ↔ API)
- Muhim kashfiyotlar — 10 ta gotcha
- Ochiq masalalar
- Dizayn tizimi
- Bosqichlar 0–12 (step by step)
- Fayl daraxti
- Tekshirish va ish tartibi

1. Loyiha konteksti

`api/` tayyor: Ts.ED 8 + Express 5 + Prisma + PostgreSQL, JWT + RBAC, Swagger, `typedSql` analitikasi.

`ui/` esa boshqa loyihadan qolgan — EduTest test platformasi (tests / results / subjects / questions). Uning domeni KidsLearn bilan umuman mos emas: `TEACHER` roli, `SignInInput`, `TestAttempts`, `/signin` endpoint'i. Bularning hech biri yangi API'da yo'q. Ayni paytda `app/layout.tsx` `next-themes` provider'siz `useTheme` ishlatadi va `authSlice` EduTest shaklidagi `user.fullName` ni kutadi.

Shuning uchun UI noldan quriladi, lekin quyidagilar saqlanadi:

Saqlanadi	Sabab
<code>components/ui/*</code> — 17 shadcn komponent	Dizayn tizimi tayyor (base-ui uslubi)
<code>app/globals.css</code>	oklch token'lar, dark mode, <code>--chart-1..5</code> tayyor
<code>lib/utils.ts</code> (<code>cn</code>)	O'zgarishsiz
<code>store/</code> skeleti	RTK Query shu yerga ulanadi
<code>components/pagination.tsx</code>	Faqat i18n qo'shiladi
<code>api/generate-types.ts</code>	Tipni API bilan sinxron ushlaydi

2. Texnologiyalar to'plami

2.1 Mavjud (o'zgarmaydi)

Qatlam	Texnologiya	Izoh
Framework	Next.js 16 — App Router	Route group'lar bilan 4 ta layout: auth / parent / child / admin
UI kutubxonasi	React 19.2	
Til	TypeScript 5	<code>strict</code>
Styling	Tailwind CSS v4	<code>@theme inline</code> + oklch token'lar, CSS-first konfiguratsiya
Komponentlar	shadcn/ui (<code>base-nova</code>)	<code>@base-ui/react</code> ustida, Radix emas
Ikonkalar	lucide-react	Faqat UI chrome uchun
Toast	sonner	
Tema	next-themes	O'rnatilgan, lekin ulanmagan — 4-bosqichda ulanadi
Cookie	js-cookie	Ikki tokenli sessiya uchun

2.2 Qo'shiladi

```
npm install react-hook-form @hookform/resolvers zod recharts jspdf date-fns
```

Texnologiya	Nima uchun
RTK Query (<code>@reduxjs/toolkit</code> ichida)	Yangi dependency emas — RTK allaqachon bor. ~25 ta ro'yxat sahifasida cache, loading/error, invalidation avtomatik. Mutation'dan keyin ro'yxat o'zi yangilanadi.
axios (mavjud, qoladi)	RTK Query <code>fetchBaseQuery</code> o'rniga axios baseQuery bilan ishlaydi — HTTP yo'li baribir bitta. Interceptor'lar tanish naqshda qoladi va S3 ga yuklashda <code>onUploadProgress</code> ishlaydi (<code>fetch</code> buni qo'llab-quvvatlamaydi).
react-hook-form + @hookform/resolvers	~15 ta forma. Uncontrolled — har harfda re-render yo'q.
zod ^3.23.8	API input schema'lari zod'da yozilgan — frontendda takrorlanadi. Major versiya bir xil bo'lsin (v4 API'si boshqacha).
recharts	Ota-ona va admin grafiklari. shadcn bilan yaxshi ishlaydi.
date-fns	Sana formatlash, uz/en/ru locale. API'da ham ishlatilgan.
jspdf	PDF sertifikat (bonus).

2.3 Kutubxonasiz hal qilinadi

Vazifa	Yechim	Nega kutubxona emas
i18n (uz/en/ru)	Qo'lda: <code>lib/i18n/</code> — lug'at + Context + <code>useT()</code>	Ilova deyarli butunlay client-side. <code>next-intl</code> ning server imkoniyatlari kerak emas.
PWA	Qo'lda: <code>manifest.json</code> + <code>sw.js</code>	<code>next-pwa</code> Next 16 bilan muammoli; qo'lda yozilgani shaffofroq.
Ovoz (TTS/STT)	Web Speech API — brauzer ichida	Native, dependency yo'q.
Puzzle / Memory	CSS <code>background-position</code> + tap-to-swap	drag-and-drop kutubxonasi kichik bolalar uchun noqulay.

2.5 Backend (kontekst uchun)

Ts.ED 8 · Express 5 · Prisma 6.10 · PostgreSQL 16 · zod · JWT (`jsonwebtoken`) · bcrypt · Swagger (`@tsed/swagger`) · `typedSql` · AWS S3 (`@aws-sdk/client-s3`) · Docker

3. API audit natijasi

Topshiriqdagi har bir shart API'da bor-yo'qligi tekshirildi.

Shart	Holat	Endpoint / izoh
3 rol (Admin / Ota-ona / Bola)	bor	<code>USER_ROLE</code> = ADMIN PARENT; CHILD — token roli
Login / Logout / JWT / RBAC	bor	<code>/auth/*</code> , <code>RevokedToken</code> bilan haqiqiy logout
Ota-ona profili (ism, tel, email, parol)	bor	<code>/auth/register</code> , <code>/auth/profile</code> , <code>/auth/password</code>
Bir ota-ona → ko'p bola	bor	<code>/children</code> CRUD
Bola: ism, tug'ilgan sana, yosh, avatar	bor	Yosh saqlanmaydi — <code>birthDate</code> dan hisoblanadi
Yoshga qarab darslar	bor	<code>/lessons/for-me</code> — faqat o'z <code>ageGroup</code> i
Dars: nom, tavsif, yosh, muqova, video, audio, rasmlar	bor	<code>Lesson</code> + <code>LessonMedia[]</code>
6 ta o'yin	bor	Bitta <code>Game</code> + <code>GameItem</code> modeli, <code>GAME_TYPE</code> bilan
Progress (o'yin, dars, ball, yulduz, faollik, streak)	bor	<code>ChildStats</code> + <code>DailyActivity</code>
4 ta medal	bor	12 ta qoida <code>utils/awards.ts</code> da
Ota-ona statistikasi (bugun, hafta, oy, fanlar)	bor	<code>/dashboard/parent</code>
Admin paneli	bor	<code>/users</code> , <code>/children/paginated</code> , <code>/lessons</code> , <code>/games</code> , <code>/media</code> , <code>/dashboard/admin</code>
Search: bola ismi, ota-ona ismi, yosh, dars nomi	bor	Trigram indekslar ham qo'shilgan
Filter: yosh, fan, sana, faollik	qisman	Sana filtri faqat darslarda (<code>from</code> / <code>to</code>)
Bildirishnomalar (3 tur)	bor	<code>NO_ACTIVITY_TODAY</code> uchun <code>POST /notifications/daily-digest</code> (tashqi cron)
REST API + Swagger	bor	<code>/docs</code>
Pagination / Validation / Error handling	bor	<code>BasicSearch</code> , zod, <code>GlobalErrorFilter</code>
Docker Compose	qisman	<code>db</code> + <code>api</code> bor, <code>ui</code> qo'shilishi kerak
Bonus: Leaderboard	bor	<code>/dashboard/leaderboard</code>
Bonus: PWA / Ovoz / PDF / i18n	frontend	Sof frontend ishi
Bonus: Push Notification	yo'q	web-push/VAPID yo'q. In-app bildirishnoma bor
Bonus: AI tavsiyalar	yo'q	

Xulosa: API UI qurishga to'siq emas.

4. Muhim kashfiyotlar — 10 ta gotcha

Kodni o'qib topilgan, hujjatda yozilmagan narsalar. Har biri to'xtatib qo'yadi.

4.1 — Butun API camelCase

`20260810120000_rename_columns_to_camel_case` migratsiyasi bilan hamma narsa o'zgargan:

```
full_name → fullName      birth_date → birthDate      age_group → ageGroup
child_id  → childId        created_at → createdAt      correct_value → correctValue
points_per_correct → pointsPerCorrect  items_per_round → itemsPerRound
progress_percent → progressPercent      duration_seconds → durationSeconds
request_id → requestId      unread_only → unreadOnly    parent_id → parentId
```

`sortBy` ning standart kaliti ham endi `createdAt`. Eski nom yuborilsa zod uni tanimaydi va **filtr jimgina ishlamay qoladi**.

Shuningdek `api/types/*.ts` dagi umumiy `*_SELECT` konstantalari olib tashlangan — `select` shakllari endi servislar ichida inline yozilgan.

4.2 — API base URL noto'g'ri

`api/config/index.ts` → `apiRoot: process.env.API_ROOT || '/api/v1'`. `config/local.ts` uni override qilmaydi.

`ui/.env` hozir `http://localhost:9100/api` — **hamma so'rov 404**.

```
NEXT_PUBLIC_API_URL=http://localhost:9100/api/v1
```

`api/.env` da `CORS_ORIGIN=http://localhost:3000` bo'lsin. Hozir bu satr yo'q — default `*` ishlaydi, lekin `server.ts` da `origin: '*'` bo'lganda `credentials` majburan `false` qilinadi.

4.3 — Ikki token (sessiyaning eng nozik joyi)

Bola parol yozolmaydi. `USER_ROLE` enum'ida `CHILD` yo'q — u faqat token darajasidagi rol.

```
POST /auth/login          → PARENT/ADMIN token
POST /auth/children/:id/select → CHILD token { sub: <ota-ona id>, role: 'CHILD', childId }
```

Cookie	Nima
<code>kl_token</code>	Hozir ishlayotgan token (ota-ona yoki bola)
<code>kl_parent_token</code>	Bola rejimida "to'xtatib qo'yilgan" ota-ona tokeni
<code>kl_role</code>	<code>ADMIN</code> / <code>PARENT</code> / <code>CHILD</code> — <code>middleware.ts</code> shundan o'qiydi

`POST /auth/logout` tokenni `jti` bo'yicha bekor qiladi. Demak **bola rejimidan chiqishda**: avval bola tokeni bilan `logout`, keyin `kl_parent_token` tiklanadi (uning `jti` si boshqa — tirik qoladi).

To'liq chiqishda: bola tokeni bilan `logout` → ota-ona tokenini tiklash → yana `logout`.

`PUT /auth/password` ham tokenni bekor qiladi — parol o'zgargach majburan login.

4.4 — Bola tokeni hech qachon admin bo'lmaydi

Ota-onasi ADMIN bo'lsa ham sessiya CHILD bo'lib qoladi. Qurilma bolaning qo'lida bo'ladi.

4.5 — To'g'ri javob brauzerga yuborilmaydi

`GET /games/:id/play` → `correctValue` olib tashlangan, `options` aralashtirilgan. Tekshiruv faqat `POST /games/:id/submit` da, serverda.

OQIBATI

Har savoldan keyin darhol "to'g'ri/xato" ko'rsatib **bo'lmaydi**. Javoblar to'planadi, natija oxirida `submit` javobidagi `results[]` dan chiqadi. Bu 8-bosqichning butun dizaynini belgilaydi.

PUZZLE va MEMORY uchun `options` da bitta element bor (`{value: 'solved'}` / `{value: 'matched'}`). Yechilganda client `options[0].value` ni, taslim bo'lsa `null` yuboradi.

4.6 — Javob konverti va xato formati

```
// muvaffaqiyat
{ success: true, _message?: string, data: T }

// xato
{ success: false, _message: string, errors?: [{field, message}], requestId?: string }
```

`errors[]` zod'dan keladi, `field` nuqta bilan (`options.0.value`) — to'g'ridan-to'g'ri `react-hook-form.setError()` ga bog'lanadi.

4.7 — Sahifalash query'si ichma-ich

`api/server.ts` da `query parser = extended` ataylab yoqilgan:

```
?page=1&size=10&search=olma&sortBy[0][key]=createdAt&sortBy[0][order]=desc
```

`fetchBaseQuery` ning `params` i ichma-ich obyektini o'zi seriyalay olmaydi — `buildListParams()` helper'i `sortBy` ni qo'lda yassilaydi.

4.8 — Generated tiplar javob tipi EMAS

`types/models/Child.ts` da `birthDate: Date`, `parent: any`, `stats: any`. Lekin JSON orqali sana **string** bo'lib keladi, `parent` esa har endpointda boshqa `select` bilan keladi. Demak generated modellar — faqat maydon nomlari uchun manba; haqiqiy javob tiplari qo'lda `types/api.ts` da yoziladi.

4.9 — Yosh saqlanmaydi

`Child` da `age` ustuni yo'q — `birthDate` dan hisoblanadi. Javoblarda `withAge()` orqali `age` va `ageGroup` qo'shib keladi. Frontend yoshni o'zi hisoblamaydi.

Guruhlar: `AGE_1_2` (0–2), `AGE_3_4` (3–4), `AGE_5_7` (5+).

4.10 — S3 bor, lekin ikkita nuqsoni bilan

Endpoint	Holat
GET /s3/generate-policy	ishlaydi — brauzerdan to'g'ridan-to'g'ri S3 ga yuklash uchun presigned policy
GET /s3/file/*key	ishlaydi — signed URL ga redirect (ochiq, auth yo'q)
POST /s3/:folder/upload	ishlamaydi

Nuqson A: POST /s3/:folder/upload da @BodyParams('UploadFiles') file yozilgan, lekin loyihada multer ham, @MultipartFile ham yo'q. Express faqat json va urlencoded parse qiladi — file doim undefined.

Nuqson B: generatePolicy da papka ro'yxati ['brands', 'products', 'avatars'] — boshqa loyihadan qolgan. KidsLearn uchun faqat avatars mos; dars va o'yin fayllari rad etiladi.

Ishlaydigan oqim:

1. GET /s3/generate-policy?folder=avatars&contentType=image/png&filename=a.png
2. Brauzer → S3 ga to'g'ridan-to'g'ri multipart POST (policy maydonlari bilan)
3. POST /media { type, url, key, originalName, mimeType, size } ← registrga yozish

5. Ochiq masalalar

1. **S3 papka ro'yxati** — generatePolicy ga lessons, games, media qo'shilsinmi? Bir qatorlik tuzatish. Tuzatilsa /admin/media da to'liq fayl yuklash bo'ladi; tuzatilmasa URL kiritish qoladi.
2. **Sana filtri** — spec "Filter: Sana" deydi, hozir faqat darslarda. Bolalar va o'yinlarga ham from/to qo'shilsinmi?
3. **POST /s3/:folder/upload** — o'chirilsinmi yoki multer qo'shib tuzatilsinmi? Presigned policy oqimi baribir yaxshiroq (fayl API serveridan o'tmaydi).

Bular UI ishini to'xtatmaydi — 0–8 bosqichlar bemalol ketaveradi, faqat 9.6 (media) shularga bog'liq.

6. Dizayn tizimi

Uchta auditoriya: **admin** (zich, ish quroli), **ota-ona** (tinch, ma'lumotli), **bola** (ulkan, rangli). Uchta alohida CSS emas — bitta token tizimi, uchta zichlik rejimi.

6.1 Qo'shiladigan semantik token'lar

Mavjud `globals.css` yaxshi qurilgan (binafsha `--primary`, `--chart-1..5`, to'liq `.dark`). Yetishmayotgani:

```
:root {
  --success: oklch(0.62 0.16 152);  --success-foreground: oklch(0.99 0 0);
  --warning: oklch(0.70 0.16 75);   --warning-foreground: oklch(0.21 0.02 286);

  /* Yosh guruhлари — issiqdan sovuqqa: kichik → katta */
  --age-1-2: oklch(0.68 0.17 20);   /* marjon */
  --age-3-4: oklch(0.72 0.16 70);   /* qahrabo */
  --age-5-7: oklch(0.66 0.14 175);  /* firuza */

  --medal-bronze: oklch(0.62 0.11 55);
  --medal-silver: oklch(0.72 0.02 260);
  --medal-gold:   oklch(0.80 0.16 85);
  --medal-diamond: oklch(0.78 0.12 200);
}

.dark {
  --success: oklch(0.72 0.17 152);  --warning: oklch(0.80 0.16 78);
  --age-1-2: oklch(0.74 0.16 20);   --age-3-4: oklch(0.78 0.15 70);
  --age-5-7: oklch(0.72 0.13 175);
  --medal-bronze: oklch(0.70 0.10 55);  --medal-silver: oklch(0.80 0.02 260);
  --medal-gold:   oklch(0.85 0.15 85);  --medal-diamond: oklch(0.84 0.11 200);
}
```

TAILWIND V4 GOTCHA

Token'ni e'lon qilishning o'zi yetmaydi. `bg-success` utilitasi paydo bo'lishi uchun uni `@theme inline` blokida ham ro'yxatdan o'tkazish shart: `--color-success: var(--success);` Aks holda `class="bg-success"` **jimgina hech nima qilmaydi**.

6.2 Uchta zichlik rejimi

```
:root {
  /* ota-ona — qulay */
  --density-radius: 0.75rem; --density-text: 1rem;
  --density-control: 2.25rem; --density-gap: 1rem;
}

[data-mode="admin"] {
  /* zich — ko'p ma'lumot ekranga sig'sin */
  --density-radius: 0.5rem;  --density-text: 0.875rem;
  --density-control: 2rem;   --density-gap: 0.75rem;
}

[data-mode="child"] {
  /* ulkan — barmoq bilan, o'qimasdan */
  --density-radius: 1.5rem;  --density-text: 1.25rem;
  --density-control: 3.5rem; --density-gap: 1.5rem;
}
```


Layout ildiziga `data-mode` qo'yiladi. `--radius` ni almashtirmang — shadcn undan `--radius-sm/md/lg` ni hisoblaydi.

6.3 Tipografika

Kontekst	Sarlavha	Matn
Admin	<code>text-lg font-medium</code>	<code>text-sm</code>
Ota-ona	<code>text-2xl font-semibold</code>	<code>text-base</code>
Bola	<code>text-4xl md:text-5xl font-bold</code>	<code>text-xl</code>
Bola — o'yin prompt'i	<code>text-7xl md:text-9xl font-bold</code>	—

Bola rejimida **hech qachon** `text-xs` **ishlatmang**. Sarlavha shrifti: `next/font/google` dan Fredoka yoki Baloo 2 → `--font-display`.

6.4 Rang = ma'no

Rang	Faqat shu uchun
<code>primary</code> (binafsha)	Asosiy amal, faol holat, brend
<code>success</code>	Tugatilgan dars, to'g'ri javob, o'sish
<code>warning</code>	Streak xavf ostida, faollik yo'q
<code>destructive</code>	O'chirish, deaktivatsiya — xato javob emas
<code>age-*</code>	Yosh guruhi belgisi — boshqa hech joyda
<code>medal-*</code>	Faqat medal ikonkasi/ramkasi
<code>chart-1..5</code>	Faqat grafiklarda

O'yindagi javob holatlari (alohida):

- Tanlangan, tekshirilmagan → `ring-4 ring-primary` + `scale-105`
- To'g'ri (natija ekranida) → `bg-success/15 ring-success`
- Xato → `bg-warning/15 ring-warning` + "qayta urinib ko'r", **qizil emas**

Qizil rang 3 yoshli bolaga "sen xato qilding" deb baqiradi. Qahrabo "yana urinib ko'r" deydi. Bu pedagogik qaror, estetik emas.

6.5 Badge naqshi

Yosh guruhi va medal: `/15` fon + to'liq rang matn + `/30` halqa. Bu naqsh ikkala temada ham kontrastni saqlaydi, chunki hammasi bitta token'dan chiqadi.

```
AGE_1_2: 'bg-[--age-1-2]/15 text-[--age-1-2] ring-[--age-1-2]/30'
```

Qulflangan medal: `grayscale opacity-40` + qulf ikonkasi. O'chirmang — bola nimaga intilishini ko'rishi kerak.

6.6 Ikonka va rasm siyosati

Nima	Manba
UI chrome (menyu, o'q, yopish)	<code>lucide-react</code>
Kategoriya belgisi	Ma'lumotdan — <code>Category.icon</code> emoji saqlaydi
Medal belgisi	Ma'lumotdan — <code>Award.icon</code>
O'yin varianti	<code>option.image</code> bo'lsa rasm, bo'lmasa <code>option.label</code>

Seed'da rasm URL'lari **yo'q** — har bir rasm joyida fallback majburiy. Eng yaxshi fallback: kategoriya emoji'si rangli fonda, bo'sh kulrang to'rtburchak emas. Avatarsiz bolalar uchun: ism bosh harfi + `id` dan deterministik rang.

6.7 Harakat

Hodisa	Animatsiya	Davomiylik
Kartochka kirishi	<code>animate-in fade-in slide-in-from-bottom-2</code>	200ms
Dialog	<code>animate-in zoom-in-95 fade-in</code>	150ms
Javob tanlandi	<code>scale-105</code>	120ms
Yulduz paydo bo'lishi	<code>zoom-in</code> + 150ms kechikish	400ms
Medal ochilishi	<code>zoom-in spin-in</code> + konfetti	700ms
Progress bar	<code>transition-[width] ease-out</code>	500ms

Admin va ota-onada animatsiya minimal. `prefers-reduced-motion` majburiy — konfetti va spin to'liq o'chadi.

6.8 Grafik uslubi

Rang	Ma'no (butun ilova bo'ylab doimiy)
<code>--chart-1</code> binafsha	Ball (points)
<code>--chart-2</code> yashil	Tugatilgan darslar
<code>--chart-3</code> qahrabo	O'ynalgan o'yinlar
<code>--chart-4</code> qizg'ish	Faol daqiqalar
<code>--chart-5</code> ko'k	Yulduzlar

- `ResponsiveContainer` har doim; ota konteynerda aniq balandlik (`h-72`)
- Grid faqat gorizontal, `stroke="var(--border)"`
- O'q matni `fill="var(--muted-foreground)"`, `fontSize={12}`
- Sana o'qi: 7 kunda `dd MMM`, 30 kunda har 5-kun, 14 kunda har 2-kun
- Nol qiymatli kunlar tushib qolmaydi — API to'ldirib yuboradi

6.9 Layout, holat ekranlari, a11y

```
Sahifa:  p-4 md:p-6    max-w-7xl mx-auto    (admin'da cheklovsiz)
Bo'limlar: space-y-6
Grid:     grid gap-4 sm:grid-cols-2 lg:grid-cols-4    (stat kartochkalar)
Bola:     grid gap-6 grid-cols-2 md:grid-cols-3      (4 dan ko'p emas)
```

- Jadval har doim `overflow-x-auto` ichida
- Skeleton haqiqiy kontent bilan bir o'lchamda — aks holda sahifa sakraydi
- Nishon o'lchami: admin $\geq 32\text{px}$, ota-ona $\geq 40\text{px}$, **bola $\geq 64\text{px}$**
- Dark mode: faqat token — hech qachon `text-white`, `#fff`, `bg-gray-*`
- Rang yagona signal bo'lmasin — to'g'ri/xato ikonka bilan ham

7. Bosqichlar

Har bir bosqich mustaqil tekshiriladi. Keyingisiga o'tishdan oldin `npm run dev` da ishlashi kerak.

0 Tayyorgarlik va tozalash

0.1 Tiplarni generatsiya qilish

```
cd api && yarn generate:types
```

Allaqachon bajarilgan — `ui/types/` endi camelCase. API'da input o'zgargan har safar qayta ishga tushiring.

0.2 O'chiriladigan fayllar

```
app/(dashboard)/          ← butun papka
components/settings/      ← admin-subjects, admin-users, create-test-form
components/app-sidebar.tsx components/auth-gate.tsx components/user-nav.tsx
lib/services/             ← butun papka (BaseService, test/result/subject...)
store/slices/authSlice.ts ← qayta yoziladi
app/login/page.tsx app/register/page.tsx ← qayta yoziladi
types/index.ts           ← EduTest tiplari
```

`lib/axios.ts` **saqlanadi** — 2.2-bandda moslashtiriladi.

0.3 Konfiguratsiya

- `ui/.env` → `NEXT_PUBLIC_API_URL=http://localhost:9100/api/v1`
- `api/.env` → `CORS_ORIGIN=http://localhost:3000`
- `app/layout.tsx` → metadata `KidsLearn`, `next-themes` provider

Tekshiruv: `npx tsc --noEmit` — faqat "topilmadi" xatolari qolsin.

1 Javob tiplari — types/api.ts

Nima uchun qo'lda: 4.8-gotcha. Bu bosqichsiz qolgan hammasi `any` bo'lib ketadi.

```
export type ApiResponse<T> = { success: true; _message?: string; data: T }
export type ApiError = { success: false; _message: string;
  errors?: {field:string;message:string}[]; requestId?: string }
export type Paged<T> = { items: T[]; count: number }

export type AgeGroup = 'AGE_1_2' | 'AGE_3_4' | 'AGE_5_7'
export type GameType = 'COLOR_MATCH' | 'ANIMAL_SOUND' | 'LETTER_MATCH'
  | 'NUMBER_MATCH' | 'PUZZLE' | 'MEMORY'
export type MedalType = 'BRONZE' | 'SILVER' | 'GOLD' | 'DIAMOND'
export type MediaType = 'IMAGE' | 'VIDEO' | 'AUDIO'
export type ProgressStatus = 'NOT_STARTED' | 'IN_PROGRESS' | 'COMPLETED'
export type NotificationType = 'NO_ACTIVITY_TODAY' | 'NEW_LESSON' | 'AWARD_EARNED'
export type UserRole = 'ADMIN' | 'PARENT'
export type AuthRole = UserRole | 'CHILD'
```

Har bir javob shakli qaysi servisdan olinadi:

Tip	Manba
AuthMe (union)	auth.service.ts me() — PARENT: {...user, isAdmin, children[]}, CHILD: {role:'CHILD', child, parent}
ChildWithStats	child.service.ts listForParent() — +stats +age +ageGroup
ChildWithParent	pagination() — +parent{id,fullName,email,phone}
LessonListItem	lesson.service.ts pagination() select
LessonForChild	forChild() — +progress{status,progressPercent,completedAt}
LessonDetail	get() — to'liq model +category +media[] (+progress[])
GameListItem	game.service.ts pagination() select
GameForChild	forChild() — +best:{stars,score,createdAt} null
PlayRound	play() — {game, items[], count, child}
SubmitResult	submit() — {session, percent, results[], stats, awards[]}
ProgressSummary	progress.service.ts forChild()
ParentDashboard	dashboard.service.ts parent()
AdminDashboard	admin() — {totals, childrenByAgeGroup, activityChart}
LeaderboardRow	leaderboard() — {rank, points, stars, streakDays, child}
AwardsResponse	award.service.ts — {earned[], locked[], count}

Qoidalar: barcha `Date` → `string`; `_count` maydonlarini unutmang (`_count.media`, `_count.items`, `_count.children`, `_count.lessons`).

2 RTK Query — baseApi va sessiya

2.1 lib/session.ts

Sof funksiyalar, React'siz — `middleware.ts` ham shu nomlardan foydalanadi.

```
export const TOKEN_COOKIE = 'kl_token'
export const PARENT_TOKEN_COOKIE = 'kl_parent_token'
export const ROLE_COOKIE = 'kl_role'
export const LOCALE_COOKIE = 'kl_locale'

getToken() / setSession(token, role) / clearSession()
enterChildMode(childToken) // kl_token → kl_parent_token, so'ng kl_token = childToken
exitChildMode() // kl_parent_token → kl_token
hasParkedParent() // "ota-onaga qaytish" tugmasi uchun
```

Cookie `expires` — login javobidagi `expiresIn` dan. `sameSite: 'lax'`, `secure` faqat production'da.

2.2 lib/axios.ts — mavjud fayl moslashtiriladi

Hozirgi fayl deyarli to'g'ri, faqat uchta o'zgarish:

1. Token `Cookie.get('token')` emas, `getToken()` dan (yangi cookie nomi)
2. 401 da `clearSession()` + `/login` — hozirgi `logout()` dispatch o'rniga
3. Toast xabarlarini `translateServerMessage()` dan o'tadi

Mas'uliyat taqsimoti — ikki joyda takrorlanmasin:

Interceptor'da	axiosBaseQuery da
<code>Authorization</code> header biriktirish	Konvertni ochish (<code>data.data</code>)
<code>_message</code> → toast	Xatoni RTK Query shakliga o'girish
401 → sessiyani tozalash va <code>/login</code>	

2.3 store/api/baseApi.ts — axios baseQuery

`fetchBaseQuery` o'rniga axios'ni o'raydigan `BaseQueryFn`. Bu RTK Query'ning standart kengaytirish nuqtasi — cache, tag va hook'lar o'zgarishsiz ishlaydi.

```
import type { BaseQueryFn } from '@reduxjs/toolkit/query'
import type { AxiosError, AxiosRequestConfig } from 'axios'
import api from '@lib/axios'
import type { ApiError, ApiResponse } from '@types/api'

type Args = {
  url: string
  method?: AxiosRequestConfig['method']
  data?: unknown
  params?: Record<string, unknown>
}

export const axiosBaseQuery = (): BaseQueryFn<Args, unknown, ApiError> =>
  async ({ url, method = 'GET', data, params }) => {
    try {
      const response = await api<ApiResponse<unknown>>({ url, method, data, params })
      // Konvert shu yerda ochiladi – hook'lar to'g'ridan-to'g'ri payload qaytaradi,
      // har bir komponentda `.data.data` yozilmaydi.
      return { data: response.data.data }
    } catch (error) {
      const err = error as AxiosError<ApiError>
      return {
        error: err.response?.data ?? { success: false, _message: err.message },
      }
    }
  }
}
```

```
export const baseApi = createApi({
  reducerPath: 'api',
  baseQuery: axiosBaseQuery(),
  tagTypes: [ 'Auth', 'Child', 'Category', 'Lesson', 'Game', 'GameItem', 'Progress',
    'Award', 'Notification', 'Media', 'User', 'Dashboard', 'Leaderboard' ],
  endpoints: () => ({}),
})
```

`endpoints` bo'sh — har domen `injectEndpoints` bilan qo'shiladi. Sabab: fayllar domenlarga bo'linadi, lekin cache bitta store'da qoladi — `invalidatesTags` domenlar orasida ham ishlaydi (dars qo'shilsa `Dashboard` ham yangilanadi).

Server xabarlarini inglizcha (`'lesson created'`, `'saved'`). Ul uch tilli — kichik `SERVER_MESSAGES` lug'ati, mos kelmasa xom matn.

2.4 Query parametrlari

Axios'ning standart serializer'i ichma-ich obyektini **o'zi** kvadrat qavs shaklida yozadi (`sortBy[0][key]=createdAt`) — bu API kutayotgan format.

Birinci ro'yxat endpoint'ida buni bir marta tekshiring (`/lessons/paginated` ga `sortBy: [{key:'title', order:'asc'}]` yuborib, tartib haqiqatan o'zgarganini ko'ring). Ishlasa `buildListParams()` faqat bo'sh qiymatlarni tozalash uchun qoladi:

```
// undefined / '' / null tashlanadi – aks holda `?search=` bo'sh string bo'lib ketadi
buildListParams({ page, size, search, sortBy, ...filters })
```

Ishlamasa — o'sha helper'da `sortBy` ni qo'lda yassilang yoki `api.defaults.paramsSerializer` ni sozlang.

2.5 `store/store.ts`

```
reducer: { auth: authReducer, [baseApi.reducerPath]: baseApi.reducer }  
middleware: (gdm) => gdm().concat(baseApi.middleware)
```

Tekshiruv: `/health` ga query yozib `status: ok` chiqishini ko'ring.

3 Auth — slice, API, middleware, sahifalar

3.1 store/slices/authSlice.ts

```
type AuthState = {
  user: AuthUser | null // ADMIN yoki PARENT (bola rejimida – ota-onasi)
  child: ChildProfile | null // faqat CHILD sessiyasida
  role: AuthRole | null
  initialized: boolean
}
```

3.2 store/api/authApi.ts

Hook	Endpoint	Izoh
useLoginMutation	POST /auth/login	{accessToken, tokenType, expiresIn}
useRegisterMutation	POST /auth/register	Bo'sh bazada birinchi user ADMIN bo'ladi
useMeQuery	GET /auth/me	Union javob
useLogoutMutation	POST /auth/logout	
useSelectChildMutation	POST /auth/children/:id/select	{...token, child}
useUpdateProfileMutation	PUT /auth/profile	invalidatesTags: ['Auth']
useUpdatePasswordMutation	PUT /auth/password	Tokenni bekor qiladi → login

3.3 components/providers/session-provider.tsx

useMeQuery → setSession yoki setChildSession. initialized bo'lguncha skeleton.

3.4 middleware.ts — rol bo'yicha yo'nalish

```
role yo'q → himoyalangan yo'l bo'lsa /login
role=CHILD → faqat /play/* (boshqasi → /play)
role=PARENT → /dashboard /children /notifications /leaderboard
               /profile /settings /select-child
role=ADMIN → yuqoridagilar + /admin/*
/login /register → role bor bo'lsa o'z bosh sahifasiga
```

kl_role cookie'dan o'qiydi — haqiqiy tekshiruv baribir API'da.

3.5 Sahifalar

- app/(auth)/login/page.tsx — muvaffaqiyatda: cookie → /auth/me → ADMIN /admin, PARENT bolalari bor /select-child, yo'q /children
- app/(auth)/register/page.tsx — fullName, email, phone?, password
- app/(auth)/layout.tsx — markazlashtirilgan kartochka, til + tema tugmalari

```
Admin:    admin@kidslearn.uz / password123
Ota-ona:  ota-ona@kidslearn.uz / password123
```

Tekshiruv: ota-ona → /select-child; admin → /admin; token'siz /dashboard → /Login.

4 i18n, tema, umumiy primitivlar

4.1 i18n

```
lib/i18n/dictionaries/{uz,en,ru}.ts    - tekis obyekt: { 'nav.dashboard': 'Bosh sahifa' }  
lib/i18n/provider.tsx                  - LocaleProvider + useT()
```

`t('nav.dashboard')`, `t('game.score', { score: 42 })`. Til `kl_locale` cookie'da. `Dictionary = typeof uz` qilib tiplang — `en.ts` da kalit tushib qolsa TS aytadi.

Guruhlar: `nav.* auth.* child.* lesson.* game.* progress.* award.* dashboard.* admin.* common.* notification.* server.*`

4.2 Tema va token'lar

1. `ThemeProvider attribute="class" defaultTheme="system"` ni `app/layout.tsx` ga
2. Yangi semantik token'lar (6.1-band) — `:root` va `.dark` ga
3. **Ularni `@theme inline` ga ham qo'shing**
4. Zichlik token'lari: `[data-mode="admin"]`, `[data-mode="child"]`
5. `components/theme-toggle.tsx`
6. Bola uchun display shrifti → `--font-display`

4.3 Yetishmayotgan shaden komponentlar

```
npx shadcn@latest add dialog alert-dialog switch progress table popover
```

4.4 `components/shared/`

`page-header` · `data-toolbar` (debounce'li qidiruv + filtr slotlari) · `empty-state` · `loading-state` · `error-state` · `confirm-dialog` · `stat-card` · `age-group-badge` · `medal-badge` · `stars` · mavjud `pagination`
+ `hooks/use-debounced-value.ts` (300ms) — qidiruv har harfda so'rov yubormasin.

5 Bola profilini tanlash — `/select-child`

Netflix uslubi. Ota-ona kirgach shu yerga tushadi.

- `GET /children` → `ChildWithStats[]`
- Kartochka: avatar (yo'q bo'lsa emoji/inicial), ism, yosh, streak, ball
- Bosilganda → `useSelectChildMutation` → `enterChildMode(token)` → `/play`
- "+" kartochkasi → bola qo'shish dialogi
- Pastda: "Ota-ona kabineti" va "Chiqish"

Tekshiruv: tanlagach `kl_role=CHILD`; `/dashboard` ga kirsangiz middleware `/play` ga qaytaradi.

6 Ota-ona kabineti

`app/(parent)/layout.tsx` — sidebar + header (bildirishnoma qo'ng'irog'i, til, tema, profil). ADMIN bo'lsa qo'shimcha "Admin panel" havolasi.

6.1 `/dashboard`

`GET /dashboard/parent?childId=` — berilmasa birinchi farzand.

Blok	Manba
4 stat kartochka: bugungi ball, o'yinlar, darslar, faol daqiqa	<code>today</code>
Streak + jami ball + jami yulduz	<code>totals</code>
Haftalik (7 kun) / oylik (30 kun) grafik	<code>weekly.days</code> / <code>monthly.days</code>
Eng yaxshi fanlar (3)	<code>bestSubjects</code> — <code>accuracy</code> bilan
Qiyin mavzular (3)	<code>hardSubjects</code>
Oxirgi mukofotlar / o'yinlar	<code>recentAwards</code> / <code>recentSessions</code>

Grafiklar: `recharts` `AreaChart` / `BarChart`, `Tabs` bilan hafta↔oy. Ranglar faqat `var(--chart-1..5)`.

6.2 `/children`

`GET /children`, `POST /children` (`fullName`, `birthDate`, `avatar` emoji), `PUT` / `DELETE /children/:id`. Har kartochkada "Kirish" → bola rejimi.

`birthDate` validatsiyasi: kelajakda bo'lmasin, 18 yildan oshmasin.

6.3 `/children/[id]`

`GET /children/:id/progress` + `GET /children/:id/awards` (qulflangan medallar ham). Tab'lar: Umumiy / Mukofotlar / Oxirgi faollik.

6.4 `/notifications`

`GET /notifications/paginated?unreadOnly=&type=&childId=` → `{items, count, unread}`. `PUT /:id/read`, `PUT /read-all`. Header qo'ng'irog'ida `unread` (`pollingInterval: 60000`).

6.5 `/profile` va `/settings`

`PUT /auth/profile`; `PUT /auth/password` — sessiya bekor qilinadi, oldindan ogohlantiring.

7 Bola rejimi (fullscreen)

`app/(child)/layout.tsx` — `data-mode="child"`, gradient fon, sidebar yo'q. Yuqorida faqat: ism, ball, streak, "Chiqish".

- Bosiladigan har narsa $\geq 64\text{px}$, matn `text-xl` +, `text-xs` yo'q
- Bosilganda `active:scale-95`
- Har bir amalda TTS — bola o'qiy olmaydi
- Ekranda bir vaqtda 6 tadan ko'p tanlov bo'lmasin

7.1 `/play`

Uchta ulkan kartochka: Darslar · O'yinlar · Yutuqlarim. Yuqorida "Salom, {ism}!" + bugungi ball.

7.2 `/play/lessons` va `/play/lessons/[id]`

- `GET /lessons/for-me?categoryId=` — faqat o'z yosh guruhi, progress bilan
- Kategoriya chip'lari (`GET /categories`)
- Dars: `videoUrl`, `audioUrl`, `media[]` galereya
- `POST /lessons/:id/progress`: video `timeupdate` da har $\sim 10\text{s}$ → `{progressPercent, watchedSeconds}`; "Tugatdim" → `{status: 'COMPLETED'}`
- Javobda `pointsEarned` va `awards[]` → medal olsa animatsiyali dialog

Ball faqat **birinchi** tugatishda. Ikkinchi marta `pointsEarned: 0` — UI "allaqachon tugatilgan" deb ko'rsatsin.

7.3 `/play/games`

`GET /games/for-me` → har o'yinda `best:{stars,score,createdAt}|null`.

7.4 `/play/progress`

`GET /progress/me` + `GET /progress/me/awards`. Qulflangan medallar kulrang. Pastda "Sertifikat olish".

8 6 ta o'yin dvigateli

```
GET /games/:id/play → { game:{code,title,instructionAudio,pointsPerCorrect,config},
                        items:[{id, promptText, promptImage, promptAudio, options[]}],
                        count, child }
↓ javoblar client'da to'planadi
POST /games/:id/submit { answers:[{itemId, value|null}], durationSeconds }
↓
{ session:{score,stars,correctCount,...}, percent, results[], stats, awards[] }
```

`app/(child)/play/games/[id]/page.tsx` — `game.code` ga qarab dvigatel tanlaydi.

8.1 Umumiy qism — `components/games/`

Fayl	Vazifa
<code>use-game-session.ts</code>	<code>play</code> yuklash, javob to'plash, vaqt, <code>submit</code>
<code>game-shell.tsx</code>	Progress ("3 / 10"), chiqish, fon
<code>game-result.tsx</code>	Yulduz animatsiyasi, ball, "Qayta o'ynash", yangi medallar

MUHIM (4.5-GOTCHA)

Javob berilganda faqat "tanlandi" animatsiyasi. To'g'ri/xato **oxirida**, `results[]` kelgach.

8.2 `ChoiceGame` — 4 ta o'yin uchun bitta dvigatel

Kod	Prompt	Variant
<code>COLOR_MATCH</code>	<code>promptText</code> ("Qizil") + TTS	Katta rangli doira (<code>option.color</code>)
<code>ANIMAL_SOUND</code>	<code>promptAudio</code> avto-ijro + "yana eshitish"	<code>option.image</code> yoki emoji + nom
<code>LETTER_MATCH</code>	<code>promptText</code> ulkan shriftda + TTS	Harf kartochkasi
<code>NUMBER_MATCH</code>	<code>promptText</code> + shuncha nuqta	Raqam kartochkasi

`option.image` bo'lmasa `option.label` — seed'da rasm yo'q, emoji fallback foydali.

8.3 `PuzzleGame`

`promptImage` ni `config.rows × config.cols` ga bo'lish — CSS `background-position` bilan (rasmni kesish shart emas). **Tap-to-swap** — kichik bolalar uchun drag'dan oson. Yechilganda `{itemId, value: item.options[0].value}`; taslim → `null`. `promptImage` yo'q bo'lsa gradient bo'laklar fallback.

8.4 `MemoryGame`

`config.pairs` juftlik, kartochka mazmuni `options` dan (yetmasa emoji to'plamidan). Ikkitasi ochish → mos kelsa qoladi, kelmasa 800ms da yopiladi. Hammasi topilganda `{itemId, value: item.options[0].value}`.

Tekshiruv: oltalasini oxirigacha o'ynang. Birinchi o'yindan keyin `FIRST_GAME`, xatosiz o'ynasangiz `PERFECT_GAME` medali chiqadi.

9 Administrator paneli

9.1 /admin

GET /dashboard/admin: stat kartochkalar (totals), childrenByAgeGroup donut, activityChart 14 kunlik BarChart.

9.2 Ro'yxat sahifalari — bitta pattern

PageHeader (+ "Qo'shish") → DataToolBar (debounce qidiruv + filtrlar)
→ Table (skeleton → data → empty) → Pager

Birinchi /admin/users **ni yozing**, qolganlari nusxa.

Sahifa	Endpoint	Filtrlar	Qidiruv
/admin/users	GET /users/paginated	role, active	ism, email, tel, farzand ismi
/admin/children	GET /children/paginated	age, ageGroup, active, parentId	bola ismi, ota-ona ismi/emaili
/admin/lessons	GET /lessons/paginated	ageGroup, categoryId, active, from / to	nom, tavsif, fan
/admin/games	GET /games/paginated	ageGroup, code, categoryId, active	nom, tavsif, fan
/admin/media	GET /media/paginated	type	fayl nomi, key

Admin active=false yozuvlarni ko'radi, boshqalar yo'q.

9.3 /admin/categories

GET/POST/PUT/DELETE /categories, ?all=true — nafaollar bilan. name, slug (bo'sh → name dan), description, icon (emoji), color, order, active. _count.lessons / _count.games — o'chirishdan oldin ogohlantirish uchun.

9.4 /admin/lessons/[id]

title, description, categoryId, ageGroup, points, order, active, coverImage, videoUrl, audioUrl, media[].

PUT /lessons/:id da media yuborilsa **to'liq almashtiriladi**. Bittalab qo'shish uchun POST /lessons/:id/media.

Dars yaratilganda mos yoshdagi bolalarning ota-onalariga avtomatik bildirishnoma ketadi.

9.5 /admin/games/[id] — eng murakkab

- code, title, categoryId, ageGroup, pointsPerCorrect, order, active
- config — **kodga qarab boshqa forma**, xom JSON emas: PUZZLE → rows, cols · MEMORY → pairs · hammasi → itemsPerRound

- `items[]` (`GET/POST/PUT/DELETE /games/:id/items`): `promptText`, `promptImage`, `promptAudio`, `options[]` dinamik massiv (`useFieldArray`), `correctValue` — `options` dan `Select`, erkin matn emas (API `refine` bilan tekshiradi)

9.6 /admin/media

`GET /media/paginated?type=`, `DELETE /media/:id`, `POST /media`. Yuklash oqimi 4.10-gotchada.

`generatePolicy` papka ro'yxati tuzatilmaguncha faqat `avatars` ishlaydi — dars/o'yin media'si uchun URL kiritish maydoni qoladi.

10 Bonus imkoniyatlar

10.1 Leaderboard

`GET /dashboard/leaderboard?period=week|month|all&ageGroup=&limit=`. Podium (1-2-3) + jadval, Tab'lar. Bola rejimida ham (`/play/leaderboard`) — o'zi ajratiladi.

10.2 PWA

`public/manifest.json` (192/512 ikonka, `display:standalone`) + `public/sw.js` (statik: `cache-first`, `/api/*`: `network-first`) + register qiluvchi client komponent.

`Authorization` header'li javoblarni **cache qilmang** — boshqa bolaning ma'lumoti ko'rinib qolishi mumkin.

10.3 Ovoz

(a) **TTS** (`lib/speech.ts`, `speechSynthesis`) — bonus emas, **asosiy funksiya**: o'yin savoli, variant nomlari, natija, dars sarlavhasi. O'zbek ovozi kam qo'llab-quvvatlanadi → `uz` → `ru` → `en` fallback.

(b) **STT** (`SpeechRecognition`) — faqat `ChoiceGame` da, mikrofon tugmasi ixtiyoriy. Brauzer qo'llab-quvvatlamasa tugma ko'rinmasin.

10.4 PDF sertifikat

`jspdf` — ism, ball, yulduz, medallar, sana, ramka.

jsPDF standart shriftlari ' (o'zbek apostrofi) ni chiza olmaydi. Yo `addFont` bilan Unicode shrift, yo matnni ASCII-xavfsiz qiling (`o'` → `o`). Sodda yo'l — ikkinchisi.

11 Docker Compose va README

11.1 ui/Dockerfile

`next.config.ts` ga `output: 'standalone'`. Multi-stage: `deps` → `builder` → `runner` (node:20-alpine).

`NEXT_PUBLIC_API_URL` **build vaqtida** kerak (client bundle'ga kiradi) — `ARG` + `ENV` orqali beriladi, runtime `environment` yetmaydi.

11.2 Ildizda `docker-compose.yml`

`api/docker-compose.yml` ni ildizga ko'chirib `ui` qo'shiladi: `db` → `api` → `ui`.

`NEXT_PUBLIC_API_URL` **brauzer**dan ko'rinadigan manzil (`http://localhost:9100/api/v1`), konteyner ichidagi `http://api:9100` emas — so'rovni brauzer yuboradi.

`api` konteyneri `CORS_ORIGIN=http://localhost:3000` bilan.

11.3 README

Ildizdagi `README.md` hozir bo'sh (2 bayt): tavsif, arxitektura, `docker compose up`, `.env` namunalari, demo login'lar, `/docs` havolasi, rollar jadvali, qolgan ishlar.

12 Tekshirish

```
cd ui && npx tsc --noEmit && npm run lint && npm run build
```

Uchidan-uchiga ssenariy

1. `cd api && yarn db:seed && yarn dev` → `http://localhost:9100/docs`
2. `cd ui && npm run dev`
3. `ota-ona@kidslearn.uz / password123` → `/select-child`
4. Bola qo'shish (3 yosh) → tanlash → `/play`
5. Dars → tugatish → medal dialogi
6. O'yin → yulduzlar → medal
7. "Ota-onaga qaytish" → `/dashboard` → bugungi ball > 0, grafikda nuqta
8. Bildirishnomalar → mukofot xabari
9. `admin@kidslearn.uz` → `/admin` → yangi dars
10. Ota-onaga qaytish → "Yangi dars qo'shildi" bildirishnomasi
11. Dark mode, til almashtirish, 375px mobil

Har bosqichdagi sifat talablari

- **Loading:** skeleton (`isLoading`), spinner emas — haqiqiy kontent bilan bir o'lchamda
- **Error:** `isError` da `error-state` + qayta urinish
- **Empty:** `count === 0` da `empty-state`
- **Validation:** zod + maydon ostida xato + server `errors[]` ni `setError` ga bog'lash
- **Responsive:** 375px dan; jadvallar `overflow-x-auto` ichida
- **Dark mode:** faqat token
- **A11y:** `aria-label`, `alt`, fokus halqasi, `prefers-reduced-motion`

Dizayn tekshiruvi (har ekran uchun)

1. Ikkala temada oching — badge fonlari, chart tooltip, disabled tugmalar
2. 375px ga toraytiring — gorizontal scroll faqat jadvalda
3. Bola ekranlarida eng kichik bosiladigan element 64px dan katta
4. `prefers-reduced-motion: reduce` bilan — konfetti va spin o'chgan

8. Fayl daraxti (yakuniy)

```
ui/
├─ app/
│   ├── (auth)/          login, register, layout
│   ├── (parent)/        dashboard, children/[id], notifications,
│   │                   leaderboard, profile, settings
│   ├── (child)/play/     page, lessons/[id], games/[id], progress, leaderboard
│   ├── (admin)/admin/    page, users, children, categories,
│   │                   lessons/[id], games/[id], media
│   ├── select-child/
│   └─ layout.tsx, globals.css
├─ components/
│   ├── ui/               shadcn (17 mavjud + 6 yangi)
│   ├── shared/           page-header, data-toolbar, empty/loading/error-state,
│   │                   confirm-dialog, stat-card, age-group-badge,
│   │                   medal-badge, stars
│   ├── charts/           activity-chart, category-chart, age-group-chart
│   ├── games/            game-shell, choice-game, puzzle-game, memory-game,
│   │                   game-result, use-game-session
│   ├── parent/ child/ admin/
│   └─ providers/         session, locale, theme, pwa
├─ store/
│   ├── api/              baseApi + 11 domen api
│   └─ slices/            authSlice
├─ lib/
│   ├── session.ts, query-params.ts, speech.ts, certificate.ts, utils.ts
│   └─ i18n/              provider, dictionaries/{uz,en,ru}
├─ types/
│   ├── api.ts            ← qo'lda
│   └─ input/ models/ output/ ← generatsiya, tegilmaydi
├─ hooks/                 use-mobile, use-debounced-value
├─ public/                manifest.json, sw.js, icons
└─ middleware.ts, Dockerfile, .env
```

9. Ish tartibi

0 → 1 → 2 → 3 tugagach ilova "tirik" bo'ladi (kirish ishlaydi). Keyingi har bir bosqich mustaqil qo'shimcha.

Vaqt cheklangan bo'lsa ustuvorlik:

1. **0-3** — majburiy poydevor
2. **4** ning tema + primitivlar qismi
3. **5, 7, 8** — bola tajribasi, loyihaning yuragi
4. **6** — ota-ona kabineti + grafiklar
5. **9** — admin paneli
6. **4** ning i18n dictionary'si va **10** — bonus
7. **11-12**

Kalitlarni boshidan `t('...')` bilan yozib boring, tarjimalarni keyin to'ldirsangiz ham bo'ladi.